# PF Lab 7 Tasks (7)

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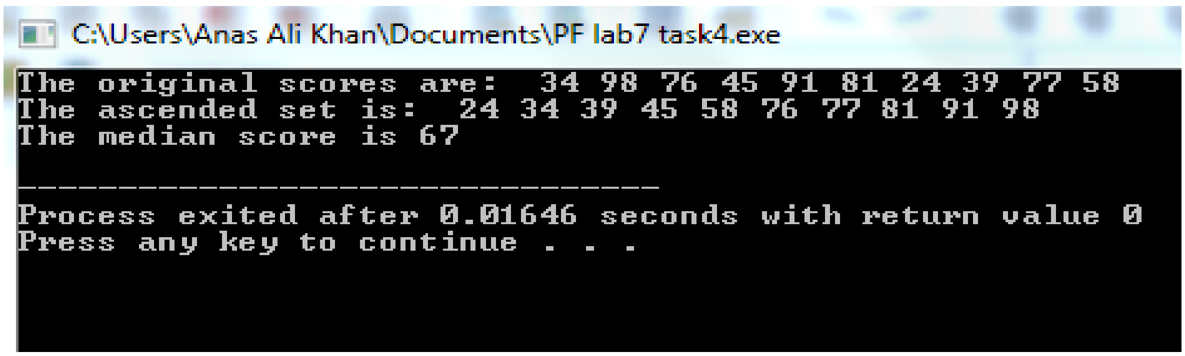
# Task 4

```
#include<stdio.h>
int main()
{
	int temp, median;
	int scores[10]={34,98,76,45,91,81,24,39,77,58};
	printf("The original scores are: ");
	for(int i=0;i<10;i++){
		printf(" %d",scores[i]);
	}

	for(int i=0;i<10;i++){
		for(int j=0;j<10-i-1;j++){
		if(scores[j]>scores[j+1]){
			temp=scores[j];
			scores[j]=scores[j+1];
			scores[j+1]=temp;
		}
		}

	}
	printf("\nThe ascended set is: ");
	for(int i=0;i<10;i++){
			printf(" %d",scores[i]);
  }
```

```
        median=(scores[4]+scores[5])/2;

        printf("\nThe median score is %d\n",median);

        return 0;


}
```




## **Task 5**

```
#include<stdio.h>

int main()

{

        int grade[10];

        float avg_grade;

        int h_grade, l_grade;

        int h_index, l_index;

        int sum=0;


        printf("Enter grades for each student\n");

        for(int i=0;i<10;i++){

                scanf("%d",&grade[i]);

        }
```

```
printf("The grades of students are:\n");
for(int j=0;j<10;j++){
	printf("Roll no %d: %d\n",j,grade[j]);
	sum+=grade[j];
}

h_grade=grade[0];
l_grade=grade[0];
for(int i=0;i<10;i++){
if(grade[i]>h_grade){
	h_grade=grade[i];
	h_index=i;
	}

	if(grade[i]<l_grade){
	l_grade=grade[i];
	l_index=i;
}

}

avg_grade=sum/10;
printf("The average grade for the class is %.2f\n",avg_grade);
printf("The highest grade in the class is %d of roll number %d\n",h_grade,h_index);
```

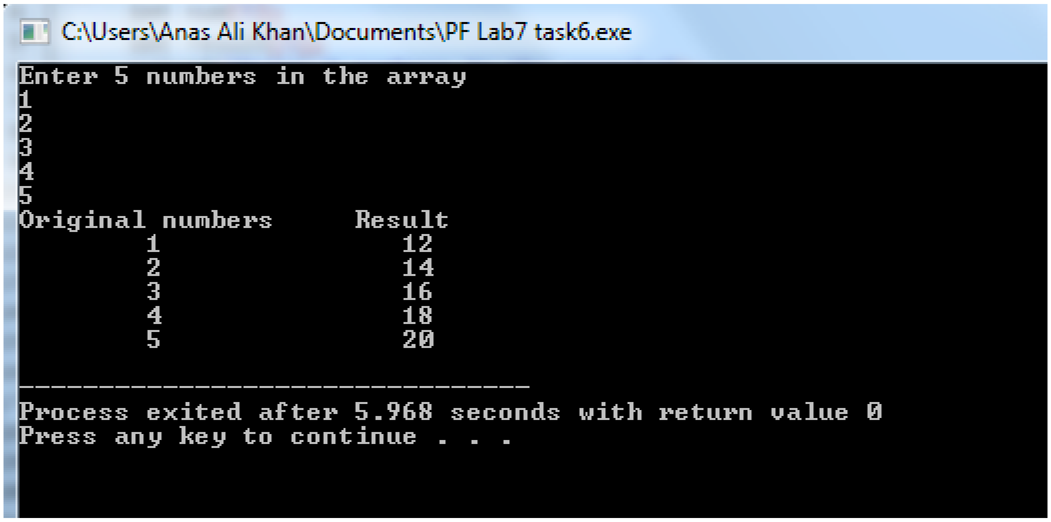
```c
	printf("The lowest grade in the class is %d of roll number %d\n",l_grade, l_index);


	printf("\nModify the score for the index with lowest score\n");
	for(int k=0;k<10;k++){
		if(k==l_index){
			printf("Enter the new score:");
			scanf("%d",&grade[k]);
		}
	}


	printf("\nThe updated grades of students are:\n");
	for(int j=0;j<10;j++){
		printf("Roll no %d: %d\n",j,grade[j]);
	}


	return 0;
}
```

C:\Users\Anas Ali Khan\Documents\PF Lab7 task5.exe

```
Enter grades for each student
45
67
78
12
93
56
61
34
85
77
The grades of students are:
Roll no 0: 45
Roll no 1: 67
Roll no 2: 78
Roll no 3: 12
Roll no 4: 93
Roll no 5: 56
Roll no 6: 61
Roll no 7: 34
Roll no 8: 85
Roll no 9: 77
The average grade for the class is 60.00
The highest grade in the class is 93 of roll number 4
The lowest grade in the class is 12 of roll number 3

Modify the score for the index with lowest score
Enter the new score:62

The updated grades of students are:
Roll no 0: 45
Roll no 1: 67
Roll no 2: 78
Roll no 3: 62
Roll no 4: 93
Roll no 5: 56
Roll no 6: 61
Roll no 7: 34
Roll no 8: 85
Roll no 9: 77

--------------------------------
Process exited after 76.49 seconds with return value 0
Press any key to continue . . .
```

# Task 6

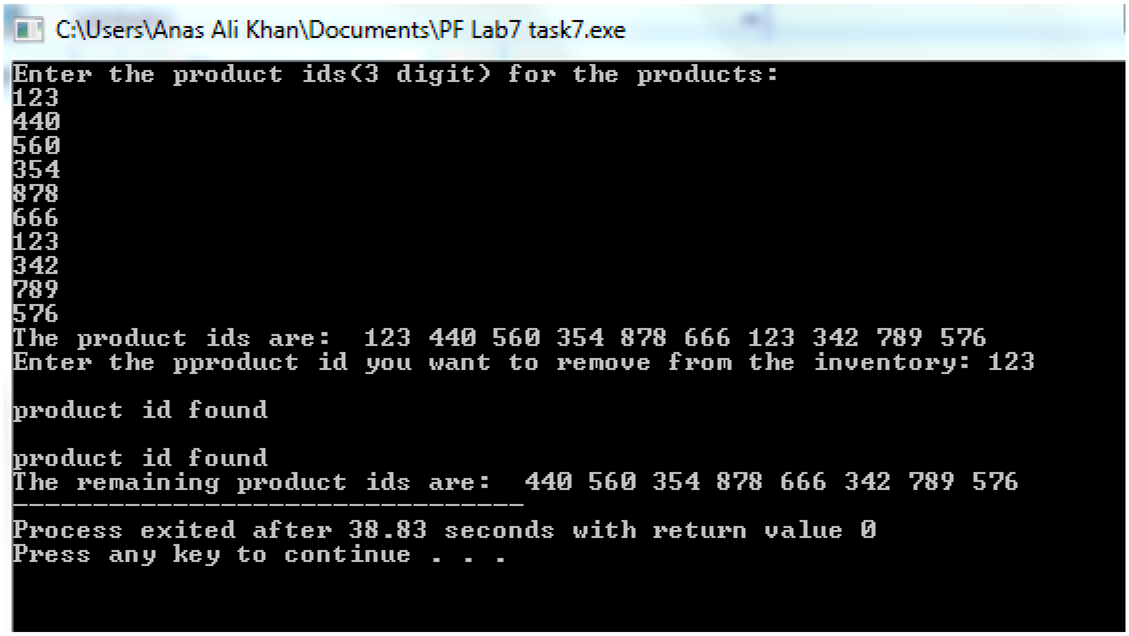
```
#include<stdio.h>

int main()

{

	int num[5];

	int result[5];

	printf("Enter 5 numbers in the array\n");

	for(int i=0;i<5;i++){

		scanf(" %d",&num[i]);

	}


	for(int j=0;j<5;j++){

		result[j]=(num[j]+10-5)*2;

	}


	printf("Original numbers     Result \n ");

	for(int i=0;i<5;i++){

		printf("\t%d\t\t%d\n",num[i], result[i]);

	}


	return 0;

}
```




## **Task 7**

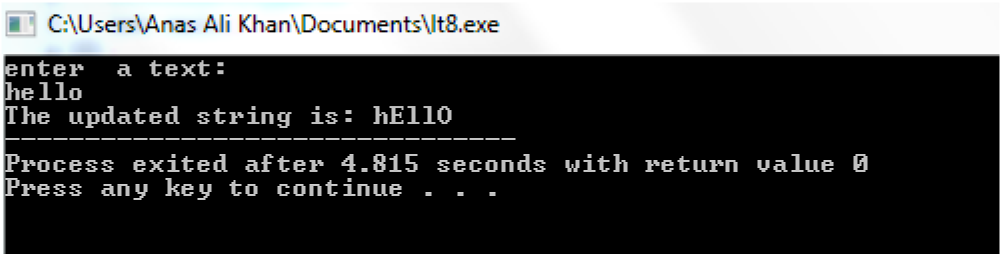
```
#include<stdio.h>

int main()

{

	int product_id[10];

	int target;


	printf("Enter the product ids(3 digit) for the products:\n");

	for(int i=0;i<10;i++){

		scanf(" %d",&product_id[i]);

	}

	printf("The product ids are: ");

	for(int i=0;i<10;i++){

		printf(" %d",product_id[i]);

	}


	printf("\nEnter the pproduct id you want to remove from the inventory: ");
```

```
	scanf("%d",&target);


	for(int j=0;j<10;j++){

		if(product_id[j]==target){

			printf("\nproduct id found\n");

			product_id[j]=0;

	}


	}


	printf("The remaining product ids are: ");

	for(int i=0;i<10;i++){

		if(product_id[i]!=0){

		printf(" %d",product_id[i]);

		}

	}


	return 0;


}
```




## Task 8

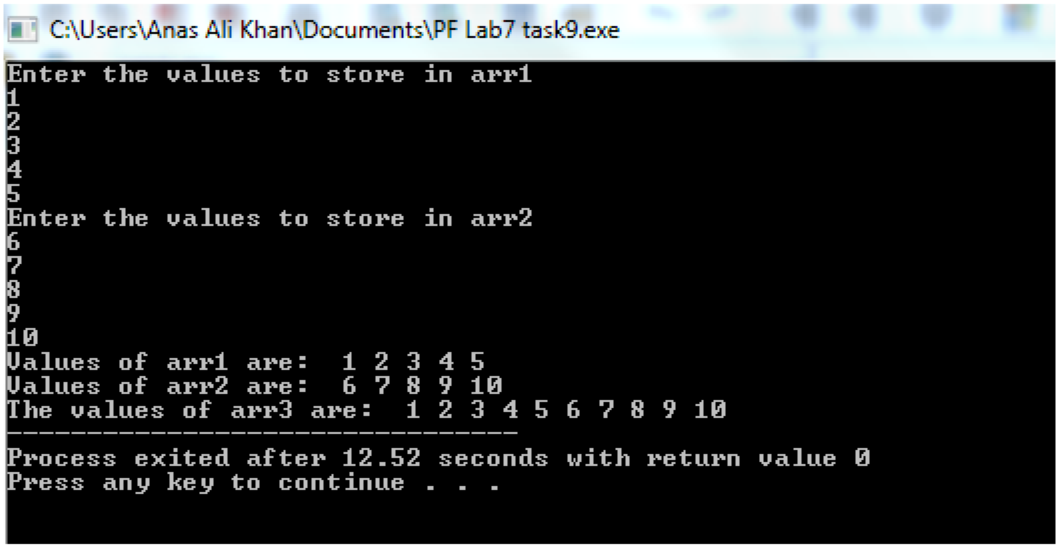
```
#include<stdio.h>

#include<ctype.h>

Int main()

{

        char string[50];

        int vowelcheck = 0;

        printf("enter  a text: \n");

        scanf("%s" , &string);

        for (int i = 0 ; i < 50 ; i++){

                if (string[i] == 'a' || string[i] == 'e' || string[i] == 'i' || string[i] == 'o' || string[i] == 'u'){

                        string[i] = toupper(string[i]);

                }

        }

        printf("The updated string is: %s" , string);

}
```




## Task 9

```
#include<stdio.h>

int main()

{

	int arr1[5];

	int arr2[5];

	int arr3[10];


	printf("Enter the values to store in arr1\n");

	for(int i=0;i<5;i++){

		scanf("%d",&arr1[i]);

	}


	printf("Enter the values to store in arr2\n");

	for(int j=0;j<5;j++){

		scanf("%d",&arr2[j]);

	}


	printf("Values of arr1 are: ");

	for(int i=0;i<5;i++){

		printf(" %d",arr1[i]);
```

```
	}

	printf("\nValues of arr2 are: ");
	for(int j=0;j<5;j++){
		printf(" %d",arr2[j]);
	}

	for(int i=0;i<5;i++){
		arr3[i]=arr1[i];
	}

	for(int j=0;j<5;j++){
		arr3[j+5]=arr2[j];
	}
	printf("\nThe values of arr3 are: ");
	for(int i=0;i<10;i++){
		printf(" %d",arr3[i]);
	}

	return 0;
}
```





## Task 10

```
#include<stdio.h>

int main()

{

	int size, element;

	printf("How many elements do you want to store in the array?\n");

	scanf("%d",&size);


	int arr[size];

	printf("Enter the values to store in the array:\n");

	for(int i=0;i<size;i++){

		scanf("%d",&arr[i]);

	}


	printf("The values in the array are:");

	for(int j=0;j<size;j++){

		printf(" %d",arr[j]);
```

```
	}

	printf("\nEnter the element you want to modify:");
	scanf(" %d",&element);

	for(int k=0;k<size;k++){
		if(k==element){
			printf("\nEnter the value you want to replace for element %d\n",element);
			scanf("%d",&arr[k]);
			break;
		}
	}

		printf("The updated array is:");
		for(int x=0;x<size;x++){
			printf(" %d",arr[x]);
		}

		return 0;
}
```

```
C:\Users\Anas Ali Khan\Documents\PF Lab7 task10.exe
How many elements do you want to store in the array?
6
Enter the values to store in the array:
2
4
6
8
10
12
The values in the array are: 2 4 6 8 10 12
Enter the element you want to modify:3

Enter the value you want to replace for element 3
9
The updated array is: 2 4 6 9 10 12
--------------------------------
Process exited after 47.36 seconds with return value 0
Press any key to continue . . .
```